DT-GSK in Plain Language

What it is, how it works, and what it proves

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A ten-page companion to the paper *DT-GSK: Dimension-Tiered Adaptive Configuration Selection and Deterministic Refinement for Gaining-Sharing Knowledge-Based Optimization Algorithm.*

For readers who already know population-based metaheuristics. Where this summary could be read as claiming more than the paper does, the paper governs.

1. Where this sits: GSK and its descendants

Gaining-Sharing Knowledge (GSK) is a population-based metaheuristic whose update is built on a two-phase human-learning metaphor. Each individual splits its coordinates into a *junior* part, which draws knowledge from nearest and farthest neighbours in the population, and a *senior* part, which draws from the best, middle and worst performers. The junior/senior split is not fixed: a decaying schedule governed by K_{exp} moves coordinates from junior to senior as the run proceeds, so the search shifts from local imitation to elite-guided refinement. Two scalars control the transfer: the knowledge factor KF (step magnitude) and the knowledge ratio KR (how many coordinates actually change); and selection is greedy.

What the descendants changed

The published variants attack the base method at three levels, and it is worth being precise about which, because it defines the gap.

- **AGSK** added the family's first adaptive parameter control: each individual draws its (KF, KR) pair from a four-setting pool whose probabilities update from realized improvements, plus a dual-regime knowledge rate and L-SHADE-style linear population reduction.
- **APGSK** extended that with a second, negative-KF pool activated after half the budget to re-diversify, a nonlinear reduction schedule, and a much larger initial population.
- **FDB-AGSK** left the parameters alone and redesigned *donor policy*: the senior-phase guide becomes fitness–distance-balance selection, aimed at AGSK's premature convergence on multimodal, hybrid and composition functions.
- **ATMALS-GSK** and **eGSK** are composite designs, adding memory-based local search and a dual adaptive knowledge factor with a gradient-based interior-point refinement respectively.

Two things are common to all of them. What adapts is a *scalar* (a parameter value, or which individual knowledge flows from), never *which coordinates move together.* And each variant fixes its behaviour at a single operating point.

The gap

Every published variant uses one configuration regardless of problem dimension. That is a real weakness, because the useful machinery changes qualitatively with D.

At $D = 10$ the space is small enough that careful structure learning and a deterministic endgame are affordable and pay off. At $D = 1000$ the same machinery consumes budget that a plain, well-scheduled population would spend better, while controllers that are pointless at low D (basin memory, subspace floors, acceptance clipping) start to matter. Evidence bases compound this: AGSK and APGSK are evaluated at $D \leq 20$ under budgets far larger relative to dimension than the CEC2017 protocol, and FDB-AGSK at a tenth of the standard budget with $D = 10$ untested.

One operating point cannot be right across that range.

2. What DT-GSK does differently

DT-GSK resolves control *by dimension* rather than at one operating point. It defines four tiers ($D < 20$, 20–49, 50–99, $D \geq 100$) and gates whole subsystems on or off across them.

The tier is resolved once, from D, before the run starts, and never changes during it. There is no online tier switching to reason about: a $D = 60$ problem simply instantiates a different machine than a $D = 15$ one.

Subsystem	$D<20$	20–49	50–99	$D\geq100$
Inherited GSK core (mask and ties modified)	on	on	on	on
NLPSR population-size reduction	on	on	on	on
ACE operator selector + ARGP pruning	on	on	on	on
Linkage block crossover[†]	on	on	on	on
Coordinate local search	on	on	on	on
BSE escape + archive, deep-stall restart	on	on	on	on
ISM interaction memory	off	off	**on**	**on**
Deterministic final polish	off	off	**on**	**on**
Upper-tier controllers	off	off	off	**on**

[†]Within the $D<20$ tier the linkage mask carries a further $D \geq 10$ gate of its own, so it is additionally inactive at the CEC2020 $D=5$ cells and at the CEC2011 problems of dimension 1, 6 and 7.

Note what this table settles: the adaptive scaffold runs at *every* tier. Three of the nine rows are gated as whole rows, and the two that most change what DT-GSK *is* above $D = 50$ (the interaction memory and the deterministic polish) do not exist below it. Only the polish is claimed as a contribution; the memory is reported as a negative result (component 8). Two further mechanisms *inside* the always-on rows carry dimension gates of their own: the ACE differential-evolution arm, which is present only for $20 \leq D < 100$, and the blockwise linkage mask, which needs $D \geq 10$. Counting those, five things switch on or off with dimension, not three.

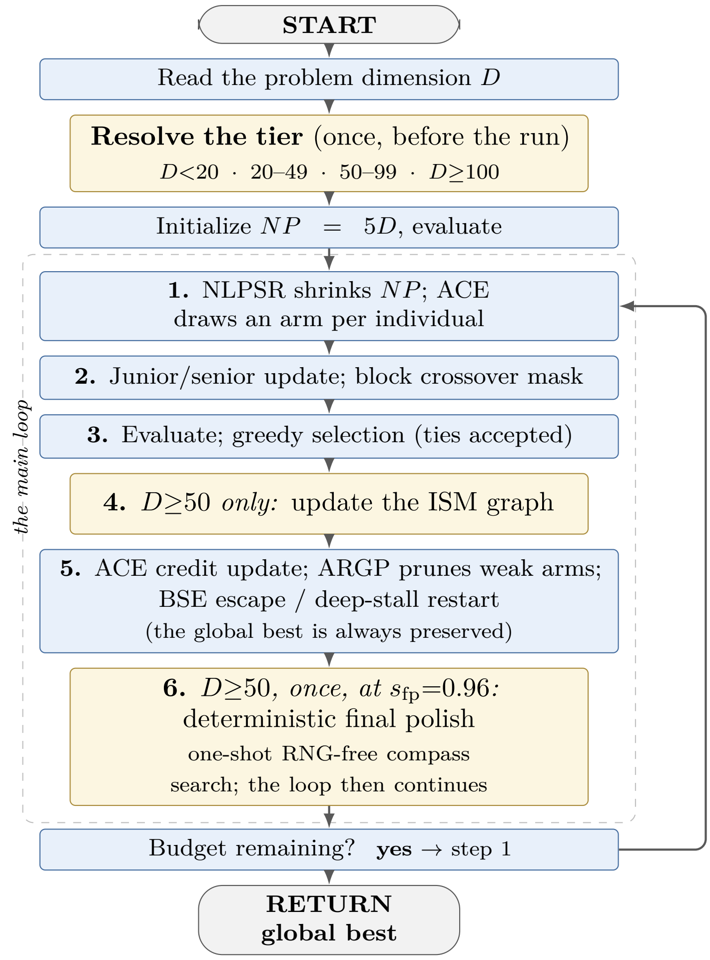

Figure 1. DT-GSK's execution order. Gold marks the dimension-gated stages; the tier itself is fixed before the loop begins. The deterministic final polish is a one-shot stage *inside* the loop, not a closing step: it arms in the first generation that reaches s_{fp}=0.96 of the budget, is capped at 0.02 MaxFES, and ordinary generations resume on the remainder.

3. The components, one by one

Nine subsystems. Six run at every tier; three are dimension-gated. Constants below are the frozen shipped configuration: no tuning is performed or permitted at run time.

1. The inherited GSK core (all tiers)
The junior/senior vector update and midpoint bound repair are numerically unchanged from base GSK. Two elements are modified: the crossover mask (see component 4) and the selection rule, which *accepts ties* rather than requiring strict improvement.

The senior partition fraction p is tier-resolved: 0.05 below $D = 20$ and 0.15 at 20–49. At $D \geq 50$ the base rate is 0.05, with the worse-ranked half widened to 0.10, but only while the trailing ten-generation mean acceptance sits at or below 0.10; above that floor the single rate 0.05 is used for the whole population. The junior dimension count follows the decaying schedule $D_{\text{jun}} = \text{clip}(\lceil D(1-x)^{K_{\text{exp}}} \rceil, 0, D)$, where x is the consumed budget fraction, so coordinates migrate from junior to senior control as the run proceeds, and at $x = 1$ every coordinate is senior.

2. NLPSR population-size reduction (all tiers)
The population starts at $NP_{\text{init}} = 5D$ and contracts on a *nonlinear* $x^{(1-x)}$ schedule toward a tier floor: $N_{\text{min}} = 12$ below $D = 50$, raised to 25 at $D \geq 50$. The alternative power-schedule exponent $\alpha_{\text{psr}} = 1.0$ is present but inert: the nonlinear form is the one shipped.

The $5D$ start is a documented fair-start exception: the six comparators run $NP = 100$ at every dimension, so at $D = 100$ DT-GSK begins with a five-fold larger population. The budget is not asymmetric: every algorithm receives the identical evaluation allowance, with initialization and every later candidate counted against it, so the larger start buys breadth at the price of generations rather than extra evaluations. **That asymmetry is now controlled, not merely disclosed.** Re-running DT-GSK at the comparators' $NP = 100$ leaves it first at 10 unknowns and second at 30, 50 and 100 (inside the top two everywhere), so the population rule is not what produces the standing. It is not innocent either: the difference is indistinguishable from zero at 10 and 30 and significant at 50 and 100, where holding the population at the panel value costs DT-GSK first place. The paper therefore qualifies its 50- and 100-unknown rank claims as resting in part on this rule.

3. ACE: adaptive configuration engine (all tiers)
A credit-based selector over a pool of five fixed arms, each a triple (KF, KR, K_{exp}) of gaining–sharing rate, crossover rate and junior-dimension exponent. Every individual independently draws an arm each generation, so a single generation exercises several configurations at once.

arm	KF	KR	K_{exp}	**character**	π_0
1	0.1	0.2	2.0	cautious explorer	0.05
2	1.0	0.1	15.0	dimension-selective bold	0.05
3	0.5	0.9	10.0	**GSK-canonical (baseline)**	**0.45**
4	1.0	0.9	5.0	aggressive wide	0.05
5	0.5	0.9	3.0	extended-exploration GSK	0.40

The pool is deliberately spread: arm 1 takes small steps on few coordinates, arm 2 large steps on very few (high K_{exp} drains the junior set fast), arm 4 large steps on most. **Arm 3 is unmodified GSK**, and with arm 5 (same rates, slower junior decay) it carries most of the initial mass, so **DT-GSK starts as base GSK and must earn any departure from it**; the three exploratory arms begin at the floor.

Two qualifications on that table, both of which change what actually executes:

- **The KR column is nominal, not executed.** A coverage floor $\min(1, \ \kappa_{\text{dim}}/D)$ is applied to every individual's KR, with $\kappa_{\text{dim}} = 0$ below $D = 20$, 20 at 20–49 and 40 at $D \geq 50$. The floor is 1.000 at $D = 20$, 0.667 at $D = 30$, 0.500 at $D = 40$, 0.800 at $D = 50$ and 0.400 at $D = 100$. So the low-KR arms never run at their listed rates above $D = 20$ (at $D = 20$ every arm updates every coordinate), and arms 1 and 2 execute at the floor.
- **π_0 is the five-arm vector.** It holds as printed at $D < 20$ and $D \geq 100$, where arms 3 and 5 carry 0.85 between them. In the two tiers that admit the sixth differential-evolution arm the five are rescaled by $1 - \pi_{\text{DE}}$ and re-projected onto the simplex, giving $(0.05, 0.05, 0.40, 0.05, 0.355, 0.095)$ at 20–49 and $(0.05, 0.05, 0.35, 0.05, 0.31, 0.19)$ at 50–99. Arms 3 and 5 then start with 0.755 and 0.660 of the mass, not 0.85.

The credit rule. Credit is an exponential moving average of realized improvement at learning rate $c = 0.10$, with probabilities projected onto the simplex under the floor $\pi_{\text{min}} = 0.05$, so no arm is ever eliminated and any arm can recover. Two details matter:

- **Warm-up.** No credit update happens until the consumed-budget fraction reaches 0.05; before that the probabilities sit at π_0.
- **Two update branches.** If a generation is *net-improving* (summed $f_{\text{old}} - f_{\text{new}}$ positive), the full-pool target is used with worsened arms floored at $\pi_{\min}$. If it yields *no net improvement*, the credit target is restricted to the arms that did improve. The restricted branch is what stabilizes learning on hard functions; without it, a generation in which everything fails would push probability mass around on noise.
- **Arm memory:** a single pool below $D = 20$; a top/bottom split at $D \geq 20$, where the individual's fitness rank selects which memory its draw uses.
- **Sixth arm:** for $20 \leq D < 100$ only, an optional differential-evolution arm ($F = 0.5$, $CR = 0.9$) is configured at 0.10 (20–49) and 0.20 (50–99); after the simplex projection it starts at 0.095 and 0.190 respectively. It is off below 20 and off again at $D \geq 100$.

4. ARGP: adaptive rank-gated pruning (all tiers)

Arms whose windowed acceptance falls below a tier threshold after warm-up are *soft-frozen* to the probability floor rather than removed, so a pruned arm recovers if it becomes productive again. Window 30 generations; threshold 0.02, tightened to 0.010 at $D \geq 50$; warm-up 0.15 of the budget.

5. Linkage block crossover (all tiers, from $D \geq 10$)

For a tier-set fraction of the population the per-coordinate KR mask is replaced by a *block* mask over non-contiguous coordinate subsets: the mechanism through which structure, rather than a scalar, is adapted. The mechanism carries its own dimension gate: it needs $D \geq 10$ below $D = 50$ and $D \geq 30$ above it, so it is inactive at the CEC2020 D=5 bank and at the CEC2011 problems of dimension 1, 6 and 7.

- **Block size:** 5 below $D = 50$, 10 at $D \geq 50$.
- **Mix probability** (the fraction of rows that get a block mask): 0.70 ($D < 20$), 0.40 (20–49), 0.70 ($D \geq 50$).
- **Refresh period:** every 20 generations below $D = 50$, every 10 at $D \geq 50$.
- **Block source:** random below $D = 50$. At $D \geq 50$ a block-mask row draws the **ISM-learned partition with probability** 0.50 and the random partition otherwise, so roughly half the block rows stay random even where the memory is active, and all of them do until the graph passes its readiness test. Nor is the crossover the memory's only route into the search: the deterministic final polish reads the signed graph directly for its basis (component 9), and at $D \geq 100$ the count of generations that commit to learned blocks releases the sample-protected population floor.

6. Charged coordinate local search (all tiers)

An axis-aligned search admitted in the endgame by the joint of a time gate, an evaluation-budget cap, a call period and a post-search cooldown. It selects the best *elite-count* individuals, orders axes by descending population standard deviation, and probes one coordinate per elite at $x_0 \pm \delta e_{\text{coord}}$, with $\delta = \mathit{step\_scale} \cdot \sigma_{\text{coord}} \cdot \mathit{step\_decay}$, where *step_decay* shrinks linearly from 1.0 to 0.25 across the endgame window. Probes are batch-evaluated and accepted greedily. Every probe is charged to the budget.

	D<20	20–49	50–99	D≥100
start fraction	0.80	0.80	0.70	0.70
eval-budget cap	0.02	0.01	0.05	0.02
elite count	2	3	5	2
step scale	0.20	0.20	0.10	0.10
call period (gens)	1	1	1	2
cooldown (gens)	6	6	6	3

The *coordinate* method is the one enabled. An ISM-block subspace variant exists in the code but is switched off in the frozen configuration, a point the paper corrects explicitly, since an earlier description had it the other way round.

7. BSE escape, archive, and the deep-stall restart (all tiers)

A stagnation escape with a *triple trigger* at $D \geq 20$ (acceptance floor 0.10, diversity floor 0.35, and a signal test with $\epsilon = 10^{-8}$ over a signal window of 10); below $D = 20$ it falls back to a single stagnation trigger with the two floors inactive. Detection window $W = 50$.

When it fires, the top round$(0.05\,NP)$ individuals by fitness are *frozen*, exempt from both rescue and restart. A Cauchy rescue then perturbs the worst round$(0.10\,NP)$ unfrozen rows by $r_s \cdot D^{-1/2} \cdot \mathrm{span} \cdot \mathcal{C}(0, 1)$, with $r_s = 0.04$ at $D \geq 20$ and 0.05 below. A rescue that lowers the incumbent *cancels* the restart for that generation; only a failed rescue triggers the archive-seeded partial restart.

- **Restart fraction:** 0.30 below $D = 50$, 0.10 at $D \geq 50$.
- **Max restarts:** 4 at $D < 20$ and 50–99; 2 otherwise, a hard cap, so BSE can never overshoot MaxFES.
- **Archive:** fixed cap 200 at all D, L2 distance filter 0.05.
- **Deep-stall restart:** fires at stall fraction 0.25 of the budget, minimum budget 20 000 evaluations, cooldown 0.15, stop at 0.9. It fully re-initializes the working population while the preserved global best can never lose ground.

8. ISM: interaction-structure memory ($D \geq 50$ *only*)

Two co-evolving accumulators (a magnitude graph G_{abs} and a signed graph G_{signed}) built from *strictly improving* accepted moves, sharing $G_\bullet \leftarrow \lambda G_\bullet + \eta \sum_i w_i\, v_i v_i^\top$ with decay $\lambda = 0.95$ and commit rate $\eta = 1.0$. Rejected trials and ties contribute nothing. Per-move source weights: 0 for differential-evolution-arm moves, 0.25 for local-search moves, 1 otherwise.

- **Gate:** *interaction-graph-min-dim* = 50. Below that tier the graph is not allocated at all.
- **Refresh:** every 5 generations below $D = 100$, every 20 at $D \geq 100$; warm-up 0.10.
- **Update thinning at $D \geq 100$:** evidence is injected only every 10th generation and capped at the 24 largest improvements. On the other nine generations the graph only decays, so the effective decay between injections is $\lambda^{10} \approx 0.60$ rather than 0.95.
- **Confidence gate:** $\kappa_{\min} = 0.35$ for linkage (0.45 for the dormant subspace search), adaptive at $D \geq 50$ on a rolling median (window 30, percentile 0.50, floor 0.12).
- **Block extraction:** max-normalize, keep row edges $\geq \max(\varphi, \frac{1}{2} \max_k W_{jk})$ with $\varphi = 10^{-4}$, symmetrize, take connected components; discard below $b_{\min} = 2$, keep whole up to $b_{\max} = 10$, split larger ones greedily.
- **Readiness:** a block feeds the crossover only after ≥ 10 updates and 2 refreshes, with at least 4 non-trivial dimensions.
- **Escape response:** on any BSE escape event the retained graph is halved ($\times 0.50$) with a 5-generation cooldown. The deep-stall restart does *not* touch the graph.

Measured cost and outcome. ISM consumes no extra objective evaluations, but the isolation measured a wall-clock overhead of **+57.3% (CEC2017 D=50), +36.3% (D=100) and +30.3% (CEC2013 D=50)**, and returned **no detectable standalone benefit at its active tiers.** That null is a registered result, advertised in the abstract and conclusions.

A later, sharper test went further. The route this test could isolate is the basis the pair-map hands the deterministic polish, and testing that route directly (no refinement, the plain coordinate axes, the learned basis) found the **learned basis is worse than the plain axes** at 50 unknowns, losing on 25 of 29 problems. The refinement itself is fine, and beats doing nothing at both sizes where it runs; what fails is the learned geometry it was given. So the pair-map is not a secondary mechanism that merely failed to help. In the route that was tested, it subtracted from something that works without it, and the paper now reports it as a specified negative result rather than as a component of the contribution.

9. The deterministic final polish ($D \geq 50$ *only*)

One-shot, RNG-free compass search, armed at budget fraction 0.96 under a cap of 0.02 MaxFES. Its basis V is the eigenvectors of the symmetrized G_{signed} ordered by descending |eigenvalue| (falling back to coordinate axes when the graph carries no signal).

Unlike the crossover and local-search consumers it is gated *only* on a non-zero signed signal, not on graph readiness, so it can act before $\kappa_{\min}$ would admit a linkage block. The step initializes at 0.05 of the box span, grows by 1.3 after an improving probe (capped at half the span), halves after a full non-improving sweep, and terminates at the budget cap or when the step falls below 10^{-12} of the span. Probes are accepted only on strict improvement, so the stage cannot corrupt the returned solution.

Repeatability follows from having no RNG draws: the same input returns bit-identical output, *in the declared single-threaded environment.* The paper is explicit that this is not a cross-thread or cross-platform guarantee.

Measured outcome: the polish works, its basis does not. A three-arm isolation at fixed enablement (no refinement, the coordinate axes, the learned eigenframe) separates the two questions. The refinement itself is confirmed: against no refinement the shipped eigenframe arm wins on 22 of 29 functions at D=50 and 23 of 29 at D=100, and the coordinate-axis variant wins 28 of 29 and 25 of 29. The *learned* basis is not: at D=50 the coordinate axes beat it on 25 of 29 ($p_{\text{Holm}} = 1.4 \times 10^{-4}$) and, under the paper's canonical tie rule, at D=100 as well ($p_{\text{Holm}} = 0.0489$). The paper therefore claims this contribution **basis-neutrally** (Supplementary S9.1): what earns its place is the deterministic, budget-exact endgame, not the geometry it searches along.

10. Upper-tier controllers ($D \geq 100$ *only*)

- **A1 late-accept clip**: tightens elitist acceptance late in the budget. τ-floor 0.85; high-acceptance 0.30; log-drop $\epsilon = -2.0$; streak $k = 5$; strict $\epsilon = 10^{-3}$; log-drop EMA $\alpha = 0.30$.
- **A2 frozen-streak broaden**: widens KR during frozen streaks. Acceptance floor 0.05; log-drop $\epsilon = -3.0$; streak $k = 10$; window/cooldown 20/10 generations; KR factor 1.5.
- **FC4 late linkage random-mix**: reverts to random linkage when measured learned-block lift turns negative. τ-floor 0.50; negative lift -0.05; severe lift -0.20; strength 0.5; lift-EMA $\alpha = 0.20$.
- **Basin memory** and a **subspace-sampling floor**: enabled.
- A **budget/ROI gate**, which admits the coordinate local search only from budget fraction 0.70, the same threshold as this tier's local-search start fraction, so it is very nearly redundant, though the two are read at different points in the generation and can disagree on the generation that crosses the threshold. The trust-region displacement governor it descends from was *removed* from the code at the paper freeze and is not present at run time.

Determinism layer

Thirteen append-only, prefix-locked RNG substreams (init, core, ace, kexp, div, bse, arch, link, de, control, flow, basin, trust). Stream 0 is the run seed verbatim; children are $s_k = (\text{seed} + 1{,}000{,}003(k{+}1)) \bmod M_{\text{seed}} + 1$ with $M_{\text{seed}} = 2{,}147{,}483{,}646$. The substream layout is prefix-locked so adding a stream cannot perturb existing ones.

4. Frozen parameter reference

Every value this summary refers to, in one place; the exhaustive set is in Supplementary S5 and the repository's algorithm freeze manifest. Tier columns are D<20 / 20–49 / 50–99 / D≥100; a single value means it does not vary by tier. The whole configuration is hash-frozen and is neither tuned nor tunable at run time.

Parameter	Value	Note
Run level		
generator	Threefry, unified shared seed, 13 substreams	prefix-locked
MaxFES	suite-dependent; $10^4 D$ on CEC2017/CEC2013	
bounds	uniform scalar (ℓ, u)	per-coordinate on CEC2011
NP_{init}	$5D$	fair-start exception
seed modulus M_{seed}	2,147,483,646	
Core GSK operator		
KF	0.5	ACE arm-controlled
KR	0.9	ACE arm-controlled
KR coverage floor κ_{dim}	0 / 20 / 40 / 40	floor $\min(1, \kappa_{\text{dim}}/D)$ overrides low-KR arms
K_{exp}	10	junior-dimension schedule
p (senior)	0.05 / 0.15 / 0.05 at D≥50	bottom half widened to 0.10 only while mean acceptance ≤ 0.10
NLPSR		
schedule	nonlinear $x^{(1-x)}$	
$N_{\min}$	12; 25 at D≥50	tier floor
α_{psr}	1.0	inert (nonlinear shipped)
ACE / ARGP		
arm pool	5 arms (KF, KR, K_{exp}); arm 3 GSK-canonical $(0.5, 0.9, 10)$	see arm table, Sec. 4.3

Parameter	**Value**	**Note** *(continued)*
initial probabilities π_0	$(0.05, 0.05, 0.45, 0.05, 0.40)$ at $D<20$ and $D\geq100$; $(0.05, 0.05, 0.40, 0.05, 0.355, 0.095)$ at 20–49; $(0.05, 0.05, 0.35, 0.05, 0.31, 0.19)$ at 50–99	0.85 / 0.755 / 0.660 on the two GSK-rate arms
learning rate c	0.10	EMA on realized improvement
probability floor $\pi_{\min}$	0.05	simplex projection; no arm dies
update start	consumed-budget fraction 0.05	π held at π_0 before it
credit target	full pool if net-improving; else restricted to improving arms	stabilizes hard functions
memory mode	single ($D<20$) $\rightarrow$ top/bottom ($D\geq20$)	rank selects the memory
DE arm	$20 \leq D < 100$ only; $F = 0.5$, $CR = 0.9$	configured 0.10 / 0.20; starts at 0.095 / 0.190 after projection
ARGP window / threshold	30 gens / 0.02 (0.010 at $D\geq50$)	warm-up 0.15
Linkage block crossover		
block size	5 ($D<50$) / 10 ($D\geq50$)	
minimum dimension	10 ($D<50$) / 30 ($D\geq50$)	off entirely below it
mix probability	0.70 / 0.40 / 0.70	fraction of rows given a block mask
refresh period	20 gens ($D<50$) / 10 ($D\geq50$)	
block source	random; at $D\geq50$ ISM-learned with probability 0.50, random otherwise	*interaction-linkage-mix-prob* = 0.50
Coordinate local search		
start fraction	0.80 / 0.80 / 0.70 / 0.70	
eval-budget cap	0.02 / 0.01 / 0.05 / 0.02	of MaxFES
elite count	2 / 3 / 5 / 2	
step scale	0.20 / 0.20 / 0.10 / 0.10	$\times\,\sigma_{\text{coord}}$
step decay	$1.0 \rightarrow 0.25$ linear	across endgame window
period / cooldown	1/1/1/2 gens / 6/6/6/3 gens	
BSE / archive / deep-stall		
trigger mode	stagnation ($D<20$) / triple ($D\geq20$)	
window W / signal window	50 / 10	$\epsilon = 10^{-8}$
acceptance / diversity floor	0.10 / 0.35	inactive at $D<20$
elite freeze	top round$(0.05\,NP)$	exempt from rescue+restart
Cauchy rescue fraction r_c	worst round$(0.10\,NP)$ unfrozen	
Cauchy scale r_s	0.05 ($D<20$) / 0.04 ($D\geq20$)	$\times D^{-1/2}\cdot$span
restart fraction r_{rst}	0.30 ($D<50$) / 0.10 ($D\geq50$)	
max restarts $R_{\max}$	4 / 2 / 4 / 2	hard cap on MaxFES
archive	cap 200 (all D), L2 filter 0.05	
deep-stall	frac 0.25; min-budget 20 000	cooldown 0.15; stop 0.9
ISM interaction memory ($D\geq50$ only)		
enable gate	$D \geq 50$	not allocated below
decay λ / commit rate η	0.95 / 1.0	
source weights	0 (DE arm) / 0.25 (local search) / 1	per accepted move
refresh period / warm-up	5 gens ($D<100$) / 20 ($D\geq100$); 0.10	
update period / sample cap	every gen., uncapped (50–99); every 10th gen., top 24 ($D\geq100$)	effective decay $\lambda^{10} \approx 0.60$ at $D\geq100$
confidence gate $\kappa_{\min}$	0.35 linkage (0.45 dormant subspace)	adaptive: window 30, pct 0.50, floor 0.12
edge floor φ	10^{-4}	with $\frac{1}{2}$ row-max rule
block size range	$b_{\min} = 2$ to $b_{\max} = 10$	larger split greedily
readiness	≥ 10 updates, 2 refreshes, 4 non-trivial dims	
escape response	graph $\times\,0.50$, 5-gen cooldown	BSE events only
Deterministic final polish ($D\geq50$ only)		
arming fraction s_{fp}	0.96	budget cap 0.02 MaxFES
basis	eigenvectors of symmetrized G_{signed}	axes if no signal
step init / growth / halving	0.05 span / $\times1.3$ / $\div2$	growth capped at $\frac{1}{2}$ span
termination	budget cap or step $< 10^{-12}$ span	strict-improvement accept
$D\geq100$ controllers		
A1 late-accept clip	τ-floor 0.85; high-acc 0.30; log-drop -2.0; streak 5; strict 10^{-3}; EMA 0.30	
A2 frozen-streak broaden	acc-floor 0.05; log-drop -3.0; streak 10; window/cooldown 20/10; $KR \times 1.5$	
FC4 linkage random-mix	τ-floor 0.50; lift -0.05 / severe -0.20; strength 0.5; EMA 0.20	
basin memory, subspace floor	enabled	
budget/ROI gate	admits local search from budget fraction 0.70	duplicates this tier's local-search start; trust-region governor removed at the freeze

5. The method in pseudocode

INPUT: objective f, dimension D, bounds $[\ell, u]$, budget MaxFES (suite-dependent), seed.

1. Resolve the tier from D (once, before the run):
$D < 20$ $\to$ scaffold only (ACE+ARGP, NLPSR, blocks, LS, BSE, restart)
$20-49$ $\to$ + differential-evolution arm in the ACE pool
$50-99$ $\to$ + ISM graph, ISM-fed linkage blocks, deterministic final polish
$D \geq 100$ $\to$ + upper-tier controllers (basin memory, subspace floor, ...);
and the DE arm switches off again

2. Open the 13 RNG substreams from the run seed; **initialize** $NP \leftarrow 5D$ from the *init* substream; evaluate.

3. REPEAT while budget remains:
a. ACE selects an arm (KF, KR, K_{exp}) by EMA credit; ARGP soft-freezes under-performing arms to $\pi_{\min}$ after warm-up.
b. Partition coordinates junior/senior by the K_{exp} schedule; build the trial vector (junior: near neighbours; senior: best/middle/worst).
c. Apply the crossover mask: block mask for 70% of rows (40% at $20-49$); at $D \geq 50$ half of those rows use ISM-supplied blocks once the graph is ready, the rest stay random; below 50 all are random.
d. Repair to bounds; evaluate; greedy selection, accepting ties (tightened to strict improvement by the A1 clip late in the budget at $D \geq 100$).
e. Update ACE credit; ARGP soft-freezes under-performing arms.
f. *If* $D \geq 50$*:* update the ISM graphs from strictly improving accepted moves; re-extract linkage blocks on the tier cadence.
g. *If stagnating:* BSE Cauchy rescue on the worst rows; if it fails, archive-seeded partial restart. On a deep stall, full re-initialization (the global best is never discarded).
h. *If* $D \geq 50$*, once, on the first generation that reaches* $s_{\text{fp}} = 0.96$ *of the budget:* deterministic final polish, compass search along the signed-graph eigenbasis (step, evaluate, contract), no RNG draws, capped at 0.02 MaxFES. The loop then continues on the remaining budget.
i. NLPSR shrinks NP toward its tier floor for the next generation.

4. RETURN the global best ever accepted.

Step 1 and steps 3f/3h are where the novelty sits: the tier resolution is the organizing contribution, and the interaction memory and the deterministic final polish are subsystems written for DT-GSK rather than inherited. The base junior/senior operator is unchanged; the contribution is *where* each mechanism is allowed to act, not a new base operator.

6. How it was tested

A claim like "this method is better" is only as strong as the comparison behind it. This study was built so the comparison can be checked.

Who competed

Seven methods: DT-GSK plus **six published methods from the same family** (GSK, AGSK, APGSK, FDB-AGSK, ATMALS-GSK, and eGSK).

The six were not compared using numbers copied out of their original papers. They were re-built and re-run inside this project, so every method faced exactly the same problems and the same budget.

One disclosure the paper makes, and this summary repeats: five of the six competing methods were written or co-written by two of this paper's own authors, including the original GSK. No method from outside this family is included in any comparison.

On what

Five standard test collections that this research community uses. That came to 12,265 runs per method, or about **86,000 runs in total**.

Collection	What is in it	Sizes	Runs each
CEC2017	29 mathematical test problems	10–100 unknowns	51
CEC2011	22 real-world problem descriptions	as given	25
CEC2013	28 mathematical test problems	10–50 unknowns	51
CEC2020	10 problems, all small	5–20 unknowns	30
CEC2013LSGO	15 very large problems	1,000 (two at 905)	25

Under what rules

- **Equal budget.** Every method got exactly the same number of tests on every problem. Nobody got extra tries.
- **Matched starting points.** On each problem, every method used the same recorded random seed. The six older methods also started from an identical set of candidate answers. **DT-GSK is the one exception the paper documents**: it builds its own starting set from the same seed, because it uses a different group size. The authors flag this as the one place the comparison is not perfectly matched.
- **An analysis plan written in advance — for one collection.** For CEC2020, the authors wrote down exactly how they would analyse the results *before* running them, and saved that plan with a timestamp. That makes it impossible to pick a flattering analysis afterwards. But this covers CEC2020 only. The other four collections had no plan committed in advance, and on the very large problems the authors say they had already seen the standings before writing their plan.

The statistical protocol

Standings are **Friedman mean ranks** over the seven-member panel, taken per dimension and aggregated. Separation is tested with **Wilcoxon signed-rank** tests at the function level, **Holm-corrected** within each dimension family, with Nemenyi critical differences reported for the omnibus layer and rank-biserial effect sizes alongside.

Two disciplines matter for reading the results:

- **The rank aggregates carry no attached test.** The headline cross-dimension figures are descriptive summaries; no cross-dimension test is claimed for them.
- **Evidential tiers never cross.** An omnibus or Nemenyi separation is never reported as a paired-test result, and the CEC2013LSGO descriptive layer, inspected before the analysis plan was signed, carries no confirmatory weight anywhere.

7. What they found

Collection	What kind of test it is	DT-GSK's place	Avg. rank
CEC2017	the collection its settings were chosen on, so the least independent of the five	**1st of 7**	2.48
CEC2013	a second check, but not an independent benchmark	**1st of 7**	2.80
CEC2013LSGO	added later, exploratory; standings were looked at before the plan was written	tied 1st with AGSK	3.13
CEC2011	real-world problem descriptions	2nd, behind eGSK	3.36
CEC2020	the one test planned in advance	**4th of 7** (AGSK 1st, 2.09)	4.11

In short: first on two collections, tied first on one, second on one, and fourth on one.

Reading the good results fairly

Three things limit them.

Who it was compared against. DT-GSK was tested against its own family, not against the strongest optimization methods in the world. The paper makes no claim about the wider field. On the 1,000-unknown problems it says so plainly: specialist large-scale methods report much better published results than any method in this family, DT-GSK included.

No test is attached. The first places are average-rank summaries. Against its closest rival, eGSK, the paired tests separate the two on the main collection only at 10 unknowns; at 30, 50 and 100 they do not, and the wider all-methods comparison separates them at no size at all.

Where the settings came from. CEC2017 gave the headline first place, and CEC2017 is also the collection the authors used to choose DT-GSK's settings. Six candidate setups were compared on it before one was frozen. So that result should be read as performance on home ground, not as a fresh test.

The tied first place on the very large problems is the weakest of the three. On that collection the tests cannot tell six of the seven methods apart at all.

Reading the bad results fairly

Second place on the real-world set. On CEC2011, eGSK beat DT-GSK, and the statistical test says that gap is real. One clarification about what CEC2011 is: these are engineering problems taken from practice and turned into a standard benchmark. That shows the method has been tried on a broad range of problem types. It is not evidence of how it would perform in an actual deployment, and the paper is careful about that distinction.

Fourth place on the small problems. On CEC2020, where every problem has 20 or fewer unknowns, DT-GSK finished fourth of seven. The reason is built into the design: its two size-gated parts, the pair-map and the final polish, do not switch on until 50 unknowns. So here it competes without the machinery that wins it results higher up. Its self-tuning machinery is still running, so this is a real result about the method as it is configured, not a fluke.

The authors had written down, before running it, that they expected DT-GSK *not* to lead this collection. That same plan also required them to report the outcome as a real finding rather than explain it away.

Three confirmed losses, not one. CEC2011 is the only confirmed loss for a whole collection. But it is not the only one. On CEC2020 at 20 unknowns, the same test confirms that both AGSK and FDB-AGSK beat DT-GSK. Those two sit in the study's strongest category of evidence, because CEC2020 is the collection that was planned in advance.

The shape of the strength

The advantage is **tiered, and it is not simply "the bigger the better".**

Across the two mathematical collections covering 10 to 100 unknowns, DT-GSK leads at every size except 30, where it drops to second on CEC2017 and third on CEC2013. That dip now has a located cause rather than only a description: at 30 unknowns the settings DT-GSK chooses for 10 unknowns beat the settings it actually ships, on 20 of 29 problems. **The 30-unknown tier is mis-specified** — which is a fixable defect in one tier, not evidence against resolving settings by size, since the same experiment shows the size-resolved settings beating a one-size-fits-all alternative at 10 and at 50. On the small collection it leads at none of the four sizes. On the real-world set it is second. **Its strength is concentrated at 50 unknowns and above.**

8. Limits, and why you can check all this

What the paper does not claim

- It does not claim DT-GSK is the best optimization method available. It does not even claim DT-GSK is the best method in its own family overall. The paper reports its standing size by size rather than as a single verdict, and other family members beat it at some sizes and on the real-world set.
- It does not claim the pair-map works. The direct test came back negative, and a later test of the one route by which it could affect the answer came back worse than negative: the plain alternative beat it. Both are reported.
- It does not claim to run faster. One competitor, eGSK, could only be rebuilt using a substitute for one internal solver. That part uses some of the same budget and can change the answer it returns, so the eGSK results here belong to this rebuild rather than to the published original. That cuts both ways: eGSK is the method that beats DT-GSK on the real-world set.
- It does not claim the settings are the best possible. They were frozen and never re-tuned per problem, with one disclosed exception: two values were added for the very-large-problem collection after results on it already existed, so that result is not claimed to be blind to its own outcome.

Why the numbers can be checked

The unusual thing about this project is not the method. It is the paper trail.

- **Every run is kept**, stored in read-only frozen data sets rather than summarised away.
- **Results can be reproduced in the setup the authors describe.** The random number streams

are pinned to recorded starting seeds, so re-running DT-GSK on a matching setup gives back the same numbers, not merely similar ones. This promise is made for DT-GSK and this project's own analysis tools.

- **Every number in the paper can be traced** back to the data file it came from, and automatic checks confirm the paper and the data still agree.
- **The negative results are published too**: the component that did not work, the computing time it cost, the loss on the real-world set, the fourth place, and the two confirmed losses at 20 unknowns.

If you have thirty seconds to explain it

It is a search method for hard problems where you can test an answer but cannot work out the best one directly. Its main idea is that it picks its own machinery based on how big the problem is, like choosing a gear before you set off, instead of using one setting for everything. Above a certain size it also finishes with a careful, repeatable polish rather than a random one. It was tested against six other published methods from *its own family* (not against the strongest methods in the field, a comparison the paper does not make) across five standard test collections with equal budgets. It came first on two, tied first on one, second on one, and fourth on one. It is weakest on small problems, where its size-gated machinery is switched off by design, and it loses on the real-world set. One of its mechanisms was tested on its own, cost 30 to 57 percent more computing time, showed no measurable benefit, and (on a later, more direct test) turned out to be actively worse than the simple alternative it replaced. The authors published that result instead of leaving it out, and stopped claiming the mechanism as a contribution.

Sources. All standings are the descriptive average ranks in the project's frozen analysis files. Protocol figures and every scope statement come from the paper itself. This is a companion document and is not part of the submitted package. For the full method, the statistical tests, and the complete results, see the paper and its Supplementary Material.